

Day 4 — Mixed — One Above, One Below

Module: Multiplication Near a Power of 10 — Nikhilam · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: compute 102×98 with sign care.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See lesson-plan.md for any day-specific materials)

Vocabulary introduced today

- *Ūrdhva-tiryagbhyāṁ* (IAST: Ūrdhva-tiryagbhyāṁ; lit. ““vertically and crosswise” — the general-purpose sūtra”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **Mixed case (one above, one below).** Care with signs! Example: 102×98 .
 - $102 \rightarrow +2$ (excess), $98 \rightarrow -2$ (deficit).
 - Left part: $102 - 2 = 100$ (or $98 + 2 = 100$).
 - Right part: $(+2)(-2) = -4$.
 - Combine: $100 \mid -4$. But how to interpret -4 in the right slot?
2. **The “borrow back” trick.** Negative right part $\rightarrow$ subtract from left. $100 \mid -4$ means $10000 - 4 = 9996$.
3. **Alternative algebra check.** $102 \times 98 = (100+2)(100-2) = 100^2 - 4 = 9996$. ✓ (Difference of squares identity.)
4. **General mixed-case rule.** Cross-subtract (using the deficit sign) for left; multiply excess and deficit (negative result) for right; convert negative right to a borrow from left.
5. **Independent practice.** 5 problems mixing all three cases (both below, both above, mixed). 10 minutes.

Activity (15 min)

A scaled version of *Method Speed Race* (see [activities/](#)). Today's slice: focus on the part of the activity that reinforces Mixed — One Above, One Below.

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See `homework.md` for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 *pañcabhūta*).